Epic-FHIR — API category: Patient Workflow Scheduling (sections 30 – 30.3.1)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

30 API category: Patient Workflow Scheduling

File: site/apis/30-Patient-Workflow-Scheduling/index.html

R4 (4)

API | Method | URL template | Type
Appointment.Read (Appointments) (R4) | GET | /api/FHIR/R4/Appointment/{ID} | Industry-Standard
Appointment.Read (Scheduled Surgeries) (R4) | GET | /api/FHIR/R4/Appointment/{ID} | Industry-Standard
Appointment.Search (Appointments) (R4) | GET | /api/FHIR/R4/Appointment?date={date}&identifier={identifier}&patient={patient}&status={status}&service-category={service-category} | Industry-Standard
Appointment.Search (Scheduled Surgeries) (R4) | GET | /api/FHIR/R4/Appointment?patient={patient}&service-category={service-category} | Industry-Standard

STU3 (6)

API | Method | URL template | Type
Appointment.$book (STU3) | POST | /api/FHIR/STU3/Appointment/$book | Industry-Standard
Appointment.$find (STU3) | POST | /api/FHIR/STU3/Appointment/$find | Industry-Standard
Appointment.Read (Appointments) (STU3) | GET | /api/FHIR/STU3/Appointment/{ID} | Industry-Standard
Appointment.Search (Appointments) (STU3) | GET | /api/FHIR/STU3/Appointment?date={date}&identifier={identifier}&patient={patient}&status={status} | Industry-Standard
Schedule.Read (STU3) | GET | /api/FHIR/STU3/Schedule/{ID} | Industry-Standard
Slot.Read (STU3) | GET | /api/FHIR/STU3/Slot/{ID} | Industry-Standard

FHIR (unversioned) (1)

API | Method | URL template | Type
COVID SMART Scheduling Links | GET | /api/epic/2021/Scheduling/Utility/CovidVaccineSlotAvailability/{state_code} | Industry-Standard

30.1 Patient Workflow Scheduling: R4

File: site/apis/30-Patient-Workflow-Scheduling/1-R4/index.html

* Appointment.Read (Appointments) (R4)
* Appointment.Read (Scheduled Surgeries) (R4)
* Appointment.Search (Appointments) (R4)
* Appointment.Search (Scheduled Surgeries) (R4)

30.1.1 Appointment.Read (Appointments) (R4)

File: site/apis/30-Patient-Workflow-Scheduling/1-R4/10468-Appointment-Read-Appointments-R4.html
Epic API ID | 10468
FHIR version | R4
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Appointment/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10468

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for an Appointment resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Appointment | Resource | False | | | An instance of the R4 Appointment resource.

Appointment._status | primitiveExtension | True | | | The mapped appointment status. This extension is returned only for organizations in the Netherlands.

Appointment.appointmentType | CodeableConcept | True | | | The type of appointment.

Appointment.comment | String | True | | | The appointment notes.

Appointment.created | String | True | | | The date when the appointment was created.

Appointment.description | String | True | | | The brief description of the visit type as would be shown on a subject line in a meeting request or appointment list.

Appointment.end | instant | True | | | The appointment end time.

Appointment.id | String | False | | | The appointment FHIR ID.

Appointment.identifier | Identifier | False | Yes | | The appointment CSN.

Appointment.minutesDuration | Int64 | True | | | The estimated length of the appointment in minutes. This can be less than the duration between the start and end times if, for example, the actual time of the appointment is only an estimate.

Appointment.participant | BackboneElement | False | Yes | | Patient, providers, and locations associated with the appointment.

Appointment.participant.actor | Reference | True | | | The patient, providers, and locations associated with the appointment.

Appointment.participant.period | Period | True | | | The participation period of the actor.

Appointment.participant.required | String | True | | | Whether this participant is required to be present at the meeting. This covers a use-case where two doctors need to meet to discuss the results for a specific patient and the patient is not required to be present.

Appointment.participant.status | code | False | | | Participation status of the actor. This will either be “accepted” or “declined” based on the appointment status.

Appointment.patientInstruction | String | True | | | Detailed information and instructions for the patient.

Appointment.reasonCode | CodeableConcept | True | Yes | | Coded reason this appointment is scheduled.

Appointment.reasonReference | Reference | True | Yes | | Reason the appointment is to take place.

Appointment.serviceCategory | CodeableConcept | True | Yes | | A broad categorization of the service that is to be performed during this appointment. Value will always be “appointment.”

Appointment.serviceType | CodeableConcept | True | Yes | | The specific service that is to be performed during this appointment.

Appointment.specialty | CodeableConcept | True | Yes | | The specialty of a practitioner that would be required to perform the service requested in this appointment

Appointment.start | instant | True | | | The appointment start time.

Appointment.status | code | False | | | The appointment status.

Sample request

```
https://hostname/instance/api/FHIR/R4/Appointment/elmPBHPxEEEvVLKSSR6xGfQaaeOxoGVxtCt9FlmcwgQ03
```

Sample response


```
{
    "resourceType": "Appointment",
    "id": "elmPBHPxEEEvVLKSSR6xGfQaaeOxoGVxtCt9FlmcwgQ03",
    "identifier": [
        {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
            "value": "10000209577"
        }
    ],
    "status": "booked",
    "serviceCategory": [
        {
            "coding": [
                {
                    "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
                    "code": "appointment",
                    "display": "Appointment"
                }
            ],
            "text": "appointment"
        }
    ],
    "serviceType": [
        {
            "coding": [
                {
                    "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.808267",
                    "code": "6",
                    "display": "Office Visit"
                }
            ]
        }
    ],
    "appointmentType": {
        "coding": [
            {
                "system": "urn:oid:2.16.840.1.113883.5.4",
                "code": "AMB",
                "display": "Ambulatory"
            }
        ]
    },
    "reasonCode": [
        {
            "coding": [
                {
                    "system": "http://snomed.info/sct",
```

```
                    "code": "316931000119104"
                }
            ],
            "text": "Knee pain, right"
        }
    ],
    "reasonReference": [
        {
            "reference": "Condition/eMIcE44Y.K0BWJcJ.QI13kIddolPVrQtCRadjUBUtusXiVGaC07vvUp85vvtcpU1l3"
        }
    ],
    "description": "Visit for routine care",
    "start": "2022-04-20T20:00:00Z",
    "end": "2022-04-20T20:15:00Z",
    "minutesDuration": 15,
    "created": "2022-03-24",
    "comment": "Patient has new, persistent knee pain during daily activities.",
    "patientInstruction": "Please make sure patient arrives 5 minutes early to fill out any necessary paperwork before
    the appointment.",
    "participant": [
        {
            "actor": {
                "reference": "Patient/eNO3wqOfAltfnWMfWBQ1WmQ3",
                "display": "Doe, Jane"
            },
            "required": "required",
            "status": "accepted",
            "period": {
                "start": "2022-04-20T20:00:00Z",
                "end": "2022-04-20T20:15:00Z"
            }
        },
        {
            "actor": {
                "reference": "Practitioner/eb3d.4apzwRM33Q91Nm7wJA3",
                "display": "Amy Smith, MD"
            },
            "required": "required",
            "status": "accepted",
            "period": {
                "start": "2022-04-20T20:00:00Z",
                "end": "2022-04-20T20:15:00Z"
            }
        },
        {
            "actor": {
                "reference": "Location/ezECB8TyTDQJuuly5Byn.lQ3",
                "display": West Clinic"
            },
            "required": "required",
            "status": "accepted",
            "period": {
                "start": "2022-04-20T20:00:00Z",
                "end": "2022-04-20T20:15:00Z"
            }
        }
    ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

30.1.2 Appointment.Read (Scheduled Surgeries) (R4)

File: site/apis/30-Patient-Workflow-Scheduling/1-R4/10471-Appointment-Read-Scheduled-Surgeries-R4.html
Epic API ID | 10471
FHIR version | R4
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET

URL template | /api/FHIR/R4/Appointment/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10471

Description

This web service allows searching for scheduled surgical procedures. It includes surgical procedures scheduled in Epic using OpTime and Cupid Invasive Cardiology workflows. To search for performed procedures refer to the Procedure.Search (Surgeries) (R4) API.

Non-surgical appointments such as office visits, preprocedure consultations, and interventional radiology can be searched using the Appointment.Search (Appointments) (R4) API.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for an Appointment resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Appointment | Resource | False | | | An instance of the R4 Appointment resource.
Appointment.cancelationReason | CodeableConcept | True | | | Reason the surgery was canceled.
Appointment.comment | String | True | | | Additional comments about the surgery.
Appointment.created | String | True | | | When the surgery record was created.
Appointment.description | String | True | | | The procedure that was performed, where, when, and by whom.
Appointment.end | instant | True | | | Time when the surgery is scheduled to end.
Appointment.id | String | False | | | FHIR identifier for the Appointment resource.
Appointment.identifier | Identifier | True | Yes | | System-specified ID for the surgery record.
Appointment.minutesDuration | Int64 | True | | | The length of the surgery.
Appointment.participant | BackboneElement | False | Yes | | People and resources who participated in the surgery.
Appointment.participant.actor | Reference | True | | | Reference to the Patient, Practitioner, or Location involved in the surgery.
Appointment.participant.period | Period | True | | | Period of time the actor participated in the surgery.
Appointment.participant.required | String | False | | | Whether the participant is required to be present during the surgery.
Appointment.participant.status | code | False | | | Participation status of the actor. If the case is cancelled or entered-in-error, everyone will be declined. If the case is any other status, required participants will be accepted and non-required participants will be tentative.
Appointment.participant.type | CodeableConcept | True | Yes | | A code signifying the role of the participant in the appointment.
Appointment.patientInstruction | String | True | | | Instructions given to the patient before the surgery.
Appointment.priority | Int64 | True | | | The priority of the surgery using the iCal Standard classification. Epic Value FHIR Value High 1 Normal 5 Low 9 Blank Blank
Epic Value | FHIR Value
High | 1
Normal | 5
Low | 9
Blank | Blank
Appointment.reasonCode | CodeableConcept | True | Yes | | External codes for diagnosis records associated with the procedure.
Appointment.requestedPeriod | Period | True | Yes | | Period of time for which the surgery was requested to be scheduled.
Appointment.serviceCategory | CodeableConcept | True | Yes | | The type of appointment. “Surgery” for surgical procedures.
Appointment.specialty | CodeableConcept | True | Yes | | The specialty that performed the surgery.
Appointment.start | instant | True | | | Time when the surgery is scheduled to start.
Appointment.status | code | False | | | Status of the surgery. Epic Value FHIR Value Scheduled Booked Canceled Cancelled Not Scheduled Proposed Missing Information Proposed Voided Entered-in-error Pending Proposed Arrived Arrived Completed Fulfilled No Show No show
Epic Value | FHIR Value
Scheduled | Booked
Canceled | Cancelled
Not Scheduled | Proposed
Missing Information | Proposed
Voided | Entered-in-error
Pending | Proposed
Arrived | Arrived
Completed | Fulfilled
No Show | No show

Sample request

```
https://hostname/instance/api/FHIR/R4/Appointment/e7F0qvjzY3rO5LzUraEBD1A3
```

Sample response


```
{
  "resourceType": "Appointment",
  "id": "e7F0qvjzY3rO5LzUraEBD1A3",
  "identifier": [
```

```
  {
    "use": "usual",
    "type": {
      "text": "ORC"
    },
    "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.798267",
    "value": "418188"
  }
],
"status": "booked",
"serviceCategory": [
  {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
        "code": "surgery",
        "display": "Surgery"
      }
    ],
    "text": "surgery"
  }
],
"specialty": [
  {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.4.836982.5030",
        "code": "110",
        "display": "General"
      }
    ],
    "text": "General"
  }
],
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "196781001",
        "display": "Acute appendicitis with peritonitis (disorder)"
      }
    ],
    "text": "Acute appendicitis with peritonitis (disorder)"
  }
],
"description": "APPENDECTOMY OPEN [44950 (CPT®)] with Hanson, Steven in MMD East OR at  7:00 AM on 2/9/2022",
"start": "2022-02-08T20:00:00Z",
"end": "2022-02-08T22:00:00Z",
"minutesDuration": 120,
"created": "2022-02-02",
"comment": "Schedule at the first case of the day.\\r\\n",
"patientInstruction": "Patient should not eat or drink after midnight prior to day of surgery.\r\nPlease arrange
  travel plans to/from the hospital.  Do not forget to bring\r\nany medications you are currently taking.  Please
  leave any valuables at\r\nhome or with someone for safekeeping.\r\n",
"participant": [
  {
    "type": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
            "code": "PART",
            "display": "Participation"
          }
        ],
        "text": "PART"
      }
    ],
    "actor": {
      "reference": "Patient/eUzI-RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3",
      "display": "Bennett, Edgar"
    },
    "required": "required",
    "status": "accepted",
    "period": {
      "start": "2022-02-08T20:00:00Z",
      "end": "2022-02-08T22:00:00Z"
    }
  },
  {
    "type": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
            "code": "PPRF",
            "display": "primary performer"
          }
```

```
      ],
      "text": "PPRF"
    }
  ],
  "actor": {
    "reference": "Practitioner/eNb.8WB1163dZHCD5WSpXGQ3",
    "display": "Steven Hanson"
  },
  "required": "required",
  "status": "accepted",
  "period": {
    "start": "2022-02-08T20:15:00Z",
    "end": "2022-02-08T21:50:00Z"
  }
},
{
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
          "code": "PART",
          "display": "Participation"
        }
      ],
      "text": "PART"
    }
  ],
  "actor": {
    "reference": "Location/eNvgllF6SIj07NXjl8ww8nw3",
    "display": "Mmd East Or"
  },
  "required": "required",
  "status": "accepted",
  "period": {
    "start": "2022-02-08T20:00:00Z",
    "end": "2022-02-08T22:00:00Z"
  }
},
{
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
          "code": "SPRF",
          "display": "secondary performer"
        }
      ],
      "text": "SPRF"
    }
  ],
  "actor": {
    "reference": "Practitioner/e5QKlsvlrLviW4IcugVAxxA3",
    "display": "Circulator Anderson"
  },
  "required": "required",
  "status": "accepted",
  "period": {
    "start": "2022-02-08T20:00:00Z",
    "end": "2022-02-08T22:00:00Z"
  }
},
{
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
          "code": "SPRF",
          "display": "secondary performer"
        }
      ],
      "text": "SPRF"
    }
  ],
  "actor": {
    "reference": "Practitioner/eSLiMCyhfpHC-4zQUgTR9UQ3",
    "display": "Scrub Kennedy"
  },
  "required": "required",
  "status": "accepted",
  "period": {
    "start": "2022-02-08T20:00:00Z",
    "end": "2022-02-08T22:00:00Z"
  }
},
{
  "type": [
```

```
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "SPRF",
              "display": "secondary performer"
            }
          ],
          "text": "SPRF"
        }
      ],
      "actor": {
        "reference": "Practitioner/eUAxZ5XXMnRkePszqD0W5xA3",
        "display": "Anesthesiologist Anesthesia, MD"
      },
      "required": "required",
      "status": "accepted",
      "period": {
        "start": "2022-02-08T20:00:00Z",
        "end": "2022-02-08T22:00:00Z"
      }
    },
    {
      "type": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
              "code": "PART",
              "display": "Participation"
            }
          ],
          "text": "PART"
        }
      ],
      "actor": {
        "reference": "Location/elxd-pgW0IoEzZEZd0Ifl2Q3",
        "display": "MMD EAST OR 1"
      },
      "required": "required",
      "status": "accepted",
      "period": {
        "start": "2022-02-08T20:00:00Z",
        "end": "2022-02-08T22:00:00Z"
      }
    }
  ],
  "requestedPeriod": [
    {
      "start": "2022-02-03T21:30:00Z",
      "end": "2022-02-03T23:30:00Z"
    }
  ]
}
```


30.1.3 Appointment.Search (Appointments) (R4)

File: site/apis/30-Patient-Workflow-Scheduling/1-R4/10469-Appointment-Search-Appointments-R4.html
Epic API ID | 10469
FHIR version | R4
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Appointment?date={date}&identifier={identifier}&patient={patient}&status={status}&service-category={service-category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10469

Description

This web service allows searching for appointments and returns patient's up-to-date appointment information, such as the appointment date and time, provider, and location. Appointment types available in this API include, but are not limited to:

* Outpatient clinic appointments

* Radiology appointments

* Non-interventional cardiology appointments

* Inpatient hospital appointments occurring as part of an admission

Appointment types NOT available in this API include but are not limited to:

* Scheduled surgeries (use the Scheduled Surgeries subresource)
* Scheduled interventional cardiology visits
* Patient-submitted appointment requests that have not been scheduled
* Book Anywhere appointments scheduled at other health systems

Request parameters

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Appointment date.
identifier | String | True | | | Appointment identifier (encounter CSN).
patient | String | False | | | Reference to the Patient resource.
service-category | String | True | | | The type of appointment. Should be "appointment" if specified.
status | String | True

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
actor | String | True | | | The patient, providers, and locations associated with the appointment.
appointment-type | String | True | | | The type of appointment.
based-on | String | True | | | Unsupported
location | String | True | | | Location FHIR ID reference for actor.
part-status | String | True | | | Participation status of the actor. This will either be “accepted” or “declined” based on the appointment status.
practitioner | String | True | | | Practitioner FHIR ID reference for actor.
reason-code | String | True | | | Coded reason this appointment is scheduled.
reason-reference | String | True | | | Reason the appointment is to take place.
service-type | String | True | | | The specific service that is to be performed during this appointment.
slot | String | True | | | Unsupported
specialty | String | True | | | The specialty of a practitioner that would be required to perform the service requested in this appointment
supporting-info | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Appointment | Resource | False | | | An instance of the R4 Appointment resource.
Bundle.entry.resource.Appointment._status | primitiveExtension | True | | | The mapped appointment status. This extension is returned only for organizations in the Netherlands.
Bundle.entry.resource.Appointment.appointmentType | CodeableConcept | True | | | The type of appointment
Bundle.entry.resource.Appointment.comment | String | True | | | The appointment notes.
Bundle.entry.resource.Appointment.created | String | True | | | The date when the appointment was created.
Bundle.entry.resource.Appointment.description | String | True | | | The brief description of the visit type as would be shown on a subject line in a meeting request or appointment list.
Bundle.entry.resource.Appointment.end | instant | True | | | The appointment end time.
Bundle.entry.resource.Appointment.id | String | False | | | The appointment FHIR ID.
Bundle.entry.resource.Appointment.identifier | Identifier | True | Yes | | The appointment CSN.
Bundle.entry.resource.Appointment.minutesDuration | Int64 | True | | | The estimated length of the appointment in minutes. This can be less than the duration between the start and end times if, for example, the actual time of the appointment is only an estimate.
Bundle.entry.resource.Appointment.participant | BackboneElement | False | Yes | | Patient, providers, and locations associated with the appointment.
Bundle.entry.resource.Appointment.participant.actor | Reference | True | | | The patient, providers, and locations associated with the appointment.
Bundle.entry.resource.Appointment.participant.period | Period | True | | | The participation period of the actor.
Bundle.entry.resource.Appointment.participant.required | String | True | | | Whether this participant is required to be present at the meeting. This covers a use-case where two doctors need to meet to discuss the results for a specific patient and the patient is not required to be present.
Bundle.entry.resource.Appointment.participant.status | code | False | | | Participation status of the actor. This will either be “accepted” or “declined” based on the appointment status.
Bundle.entry.resource.Appointment.patientInstruction | String | True | | | Detailed information and instructions for the patient.
Bundle.entry.resource.Appointment.reasonCode | CodeableConcept | True | Yes | | Coded reason this appointment is scheduled.
Bundle.entry.resource.Appointment.reasonReference | Reference | True | Yes | | Reason the appointment is to take place.
Bundle.entry.resource.Appointment.serviceCategory | CodeableConcept | True | Yes | | A broad categorization of the service that is to be performed during this appointment. Value will always be “appointment.”
Bundle.entry.resource.Appointment.serviceType | CodeableConcept | True | Yes | | The specific service that is to be performed during this appointment.
Bundle.entry.resource.Appointment.specialty | CodeableConcept | True | Yes | | The specialty of a practitioner that would be required to perform the service requested in this appointment
Bundle.entry.resource.Appointment.start | instant | True | | | The appointment start time.
Bundle.entry.resource.Appointment.status | code | False | | | The appointment status.

Sample request

```
https://hostname/instance/api/FHIR/R4/Appointment?patient=eNO3wqOfAltfnWMfWBQ1WmQ3&service-category=appointment
```

Sample response

```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/R4/Appointment?patient=eNO3wqOfAltfnWMfWBQ1WmQ3&service-
    category=appointment"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/R4/Appointment/elmPBHPxEEEvVLKSSR6xGfQaaeOxoGVxtCt9FlmcwgQ03"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/R4/Appointment/elmPBHPxEEEvVLKSSR6xGfQaaeOxoGVxtCt9FlmcwgQ03",
      "resource": {
        "resourceType": "Appointment",
        "id": "elmPBHPxEEEvVLKSSR6xGfQaaeOxoGVxtCt9FlmcwgQ03",
        "identifier": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.3.698084.8",
            "value": "10000209577"
          }
        ],
        "status": "booked",
        "serviceCategory": [
          {
            "coding": [
              {
                "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
                "code": "appointment",
                "display": "Appointment"
              }
            ],
            "text": "appointment"
          }
        ],
        "serviceType": [
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.808267",
                "code": "6",
                "display": "Office Visit"
              }
            ]
          }
        ],
        "appointmentType": {
          "coding": [
            {
              "system": "urn:oid:2.16.840.1.113883.5.4",
              "code": "AMB",
              "display": "Ambulatory"
            }
          ]
        },
        "reasonCode": [
          {
            "coding": [
              {
                "system": "http://snomed.info/sct",
                "code": "316931000119104"
              }
            ],
            "text": "Knee pain, right"
          }
        ],
        "reasonReference": [
          {
            "reference": "Condition/eMIcE44Y.K0BWJcJ.QI13kIddolPVrQtCRadjUBUtusXiVGaC07vvUp85vvtcpU1l3"
          }
        ],
        "description": "Visit for routine care",
        "start": "2022-04-20T20:00:00Z",
        "end": "2022-04-20T20:15:00Z",
        "minutesDuration": 15,
        "created": "2022-03-24",
        "comment": "Patient has new, persistent knee pain during daily activities.",
        "patientInstruction": "Please make sure patient arrives 5 minutes early to fill out any necessary paperwork
    before the appointment.",
        "participant": [
          {
            "actor": {
```

```
              "reference": "Patient/eNO3wqOfAltfnWMfWBQ1WmQ3",
              "display": "Doe, Jane"
            },
            "required": "required",
            "status": "accepted",
            "period": {
              "start": "2022-04-20T20:00:00Z",
              "end": "2022-04-20T20:15:00Z"
            }
          },
          {
            "actor": {
              "reference": "Practitioner/eb3d.4apzwRM33Q91Nm7wJA3",
              "display": "Amy Smith, MD"
            },
            "required": "required",
            "status": "accepted",
            "period": {
              "start": "2022-04-20T20:00:00Z",
              "end": "2022-04-20T20:15:00Z"
            }
          },
          {
            "actor": {
              "reference": "Location/ezECB8TyTDQJuuly5Byn.lQ3",
              "display": "West Clinic"
            },
            "required": "required",
            "status": "accepted",
            "period": {
              "start": "2022-04-20T20:00:00Z",
              "end": "2022-04-20T20:15:00Z"
            }
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

30.1.4 Appointment.Search (Scheduled Surgeries) (R4)

File: site/apis/30-Patient-Workflow-Scheduling/1-R4/10478-Appointment-Search-Scheduled-Surgeries-R4.html
Epic API ID | 10478
FHIR version | R4
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
HTTP method | GET
URL template | /api/FHIR/R4/Appointment?patient={patient}&service-category={service-category}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10478

Description

This web service allows searching for scheduled surgical procedures. It includes surgical procedures scheduled in Epic using OpTime and Cupid Invasive Cardiology workflows. To search for performed procedures refer to the Procedure.Search (Surgeries) (R4) API.

Non-surgical appointments such as office visits, preprocedure consultations, and interventional radiology can be searched using the Appointment.Search (Appointments) (R4) API.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
patient | String | False | | | The patient on the scheduled surgery.
service-category | String | True | | | The type of appointment. Should be "surgery" if specified.

Request parameters (post-filtered search parameters)

Name | Type | Optional | Array | Cardinality | Description
actor | String | True | | | Reference to the Patient, Practitioner, or Location involved in the surgery.
appointment-type | String | True | | | Unsupported
based-on | String | True | | | Unsupported
date | String | True | | | Time when the surgery is scheduled to start (UTC).
identifier | String | True | | | System-specified ID for the surgery record.
location | String | True | | | Location FHIR ID reference for actor
part-status | String | True | | | Participation status of the actor. If the case is cancelled or entered-in-error, everyone will be declined. If the case is any other status, required participants will be accepted and non-required participants will be tentative.
practitioner | String | True | | | Practitioner FHIR ID reference for actor
reason-code | String | True | | | External codes for diagnosis records associated with the procedure.
reason-reference | String | True | | | Unsupported
service-type | String | True | | | Unsupported
slot | String | True | | | Unsupported
specialty | String | True | | | The specialty that performed the surgery.
status | String | True | | | Status of the surgery. Epic Value FHIR Value Scheduled Booked Canceled Cancelled Not Scheduled Proposed Missing Information Proposed Voided Entered-in-error Pending Proposed Arrived Arrived Completed Fulfilled No Show No show
Epic Value | FHIR Value
Scheduled | Booked
Canceled | Cancelled
Not Scheduled | Proposed
Missing Information | Proposed
Voided | Entered-in-error
Pending | Proposed
Arrived | Arrived
Completed | Fulfilled
No Show | No show
supporting-info | String | True | | | Unsupported

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | BackboneElement | False | Yes | | An array of entries included in this bundle.
Bundle.entry.resource | ResourceContainer | False | | | A container for a single FHIR resource.
Bundle.entry.resource.Appointment | Resource | False | | | An instance of the R4 Appointment resource.
Bundle.entry.resource.Appointment.cancelationReason | CodeableConcept | True | | | Reason the surgery was canceled.
Bundle.entry.resource.Appointment.comment | String | True | | | Additional comments about the surgery.
Bundle.entry.resource.Appointment.created | String | True | | | When the surgery record was created.
Bundle.entry.resource.Appointment.description | String | True | | | The procedure that was performed, where, when, and by whom.
Bundle.entry.resource.Appointment.end | instant | True | | | Time when the surgery is scheduled to end (UTC).
Bundle.entry.resource.Appointment.id | String | False | | | FHIR identifier for the Appointment resource.
Bundle.entry.resource.Appointment.identifier | Identifier | True | Yes | | System-specified ID for the surgery record.
Bundle.entry.resource.Appointment.minutesDuration | Int64 | True | | | The length of the surgery.
Bundle.entry.resource.Appointment.participant | BackboneElement | False | Yes | | People and resources who participated in the surgery.
Bundle.entry.resource.Appointment.participant.actor | Reference | True | | | Reference to the Patient, Practitioner, or Location involved in the surgery.
Bundle.entry.resource.Appointment.participant.period | Period | True | | | Period of time the actor participated in the surgery.
Bundle.entry.resource.Appointment.participant.required | String | False | | | Whether the participant is required to be present during the surgery.
Bundle.entry.resource.Appointment.participant.status | code | False | | | Participation status of the actor. If the case is cancelled or entered-in-error, everyone will be declined. If the case is any other status, required participants will be accepted and non-required participants will be tentative.
Bundle.entry.resource.Appointment.participant.type | CodeableConcept | True | Yes | | A code signifying the role of the participant in the appointment.
Bundle.entry.resource.Appointment.patientInstruction | String | False | | | Instructions given to the patient before the surgery.
Bundle.entry.resource.Appointment.priority | Int64 | True | | | The priority of the surgery using the iCal Standard classification. Epic Value FHIR Value High 1 Normal 5 Low 9 Blank Blank
Epic Value | FHIR Value
High | 1
Normal | 5
Low | 9
Blank | Blank
Bundle.entry.resource.Appointment.reasonCode | CodeableConcept | True | Yes | | External codes for diagnosis records associated with the procedure.
Bundle.entry.resource.Appointment.requestedPeriod | Period | True | Yes | | Period of time for which the surgery was requested to be scheduled.

Bundle.entry.resource.Appointment.serviceCategory | CodeableConcept | True | Yes | | The type of appointment. “Surgery” for surgical procedures.
Bundle.entry.resource.Appointment.specialty | CodeableConcept | True | Yes | | The specialty that performed the surgery.
Bundle.entry.resource.Appointment.start | instant | True | | | Time when the surgery is scheduled to start (UTC).
Bundle.entry.resource.Appointment.status | code | False | | | Status of the surgery. Epic Value FHIR Value Scheduled Booked Canceled Cancelled Not Scheduled Proposed Missing Information Proposed Voided Entered-in-error Pending Proposed Arrived Arrived Completed Fulfilled No Show No show
Epic Value | FHIR Value
Scheduled | Booked
Canceled | Cancelled
Not Scheduled | Proposed
Missing Information | Proposed
Voided | Entered-in-error
Pending | Proposed
Arrived | Arrived
Completed | Fulfilled
No Show | No show

Sample request

```
https://hostname/instance/api/FHIR/R4/Appointment?patient=eUzI-RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3&service-
    category=surgery
```

Sample response


```
{
    "resourceType": "Bundle",
    "type": "searchset",
    "total": 1,
    "link": [
        {
            "relation": "self",
            "url": "https://hostname/instance/api/FHIR/R4/Appointment?patient=eUzI-
    RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3&service-category=surgery"
        }
    ],
    "entry": [
        {
            "link": [
                {
                    "relation": "self",
                    "url": "https://hostname/instance/api/FHIR/R4/Appointment/e7F0qvjzY3rO5LzUraEBD1A3"
                }
            ],
            "fullUrl": "https://hostname/instance/api/FHIR/R4/Appointment/e7F0qvjzY3rO5LzUraEBD1A3",
            "resource": {
                "resourceType": "Appointment",
                "id": "e7F0qvjzY3rO5LzUraEBD1A3",
                "identifier": [
                    {
                        "use": "usual",
                        "type": {
                            "text": "ORC"
                        },
                        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.798267",
                        "value": "418188"
                    }
                ],
                "status": "booked",
                "serviceCategory": [
                    {
                        "coding": [
                            {
                                "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
                                "code": "surgery",
                                "display": "Surgery"
                            }
                        ],
                        "text": "surgery"
                    }
                ],
                "specialty": [
                    {
                        "coding": [
                            {
                                "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.4.836982.5030",
                                "code": "110",
                                "display": "General"
                            }
                        ],
                        "text": "General"
                    }
                ],
                "reasonCode": [
                    {
                        "coding": [
```

```
                    {
                        "system": "http://snomed.info/sct",
                        "code": "196781001",
                        "display": "Acute appendicitis with peritonitis (disorder)"
                    }
                ],
                "text": "Acute appendicitis with peritonitis (disorder)"
            }
        ],
        "description": "APPENDECTOMY OPEN [44950 (CPT®)] with Hanson, Steven in MMD East OR at  7:00 AM on
2/9/2022",
        "start": "2022-02-08T20:00:00Z",
        "end": "2022-02-08T22:00:00Z",
        "minutesDuration": 120,
        "created": "2022-02-02",
        "comment": "Schedule at the first case of the day.\\r\\n",
        "patientInstruction": "Patient should not eat or drink after midnight prior to day of surgery.\r\nPlease
 arrange travel plans to/from the hospital.  Do not forget to bring\r\nany medications you are currently taking.
Please leave any valuables at\r\nhome or with someone for safekeeping.\r\n",
        "participant": [
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PART",
                                "display": "Participation"
                            }
                        ],
                        "text": "PART"
                    }
                ],
                "actor": {
                    "reference": "Patient/eUzI-RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3",
                    "display": "Bennett, Edgar"
                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:00:00Z",
                    "end": "2022-02-08T22:00:00Z"
                }
            },
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PPRF",
                                "display": "primary performer"
                            }
                        ],
                        "text": "PPRF"
                    }
                ],
                "actor": {
                    "reference": "Practitioner/eNb.8WB1163dZHCD5WSpXGQ3",
                    "display": "Steven Hanson"
                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:15:00Z",
                    "end": "2022-02-08T21:50:00Z"
                }
            },
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PART",
                                "display": "Participation"
                            }
                        ],
                        "text": "PART"
                    }
                ],
                "actor": {
                    "reference": "Location/eNvgllF6SIj07NXjl8ww8nw3",
                    "display": "Mmd East Or"
                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:00:00Z",
```

```
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/e5QKlsvlrLviW4IcugVAxxA3",
            "display": "Circulator Anderson"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/eSLiMCyhfpHC-4zQUgTR9UQ3",
            "display": "Scrub Kennedy"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/eUAxZ5XXMnRkePszqD0W5xA3",
            "display": "Anesthesiologist Anesthesia, MD"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "PART",
                        "display": "Participation"
                    }
                ],
                "text": "PART"
            }
        ],
        "actor": {
```

```
                            "reference": "Location/elxd-pgW0IoEzZEZd0Ifl2Q3",
                            "display": "MMD EAST OR 1"
                        },
                        "required": "required",
                        "status": "accepted",
                        "period": {
                            "start": "2022-02-08T20:00:00Z",
                            "end": "2022-02-08T22:00:00Z"
                        }
                    }
                ],
                "requestedPeriod": [
                    {
                        "start": "2022-02-03T21:30:00Z",
                        "end": "2022-02-03T23:30:00Z"
                    }
                ]
            },
            "search": {
                "mode": "match"
            }
        }
    ]
}
{
    "resourceType": "Bundle",
    "type": "searchset",
    "total": 1,
    "link": [
        {
            "relation": "self",
            "url": "https://hostname/instance/api/FHIR/R4/Appointment?patient=eUzI-
    RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3&service-category=surgery"
        }
    ],
    "entry": [
        {
            "link": [
                {
                    "relation": "self",
                    "url": "https://hostname/instance/api/FHIR/R4/Appointment/e7F0qvjzY3rO5LzUraEBD1A3"
                }
            ],
            "fullUrl": "https://hostname/instance/api/FHIR/R4/Appointment/e7F0qvjzY3rO5LzUraEBD1A3",
            "resource": {
                "resourceType": "Appointment",
                "id": "e7F0qvjzY3rO5LzUraEBD1A3",
                "identifier": [
                    {
                        "use": "usual",
                        "type": {
                            "text": "ORC"
                        },
                        "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.2.798267",
                        "value": "418188"
                    }
                ],
                "status": "booked",
                "serviceCategory": [
                    {
                        "coding": [
                            {
                                "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
                                "code": "surgery",
                                "display": "Surgery"
                            }
                        ],
                        "text": "surgery"
                    }
                ],
                "specialty": [
                    {
                        "coding": [
                            {
                                "system": "urn:oid:1.2.840.114350.1.13.5325.1.7.4.836982.5030",
                                "code": "110",
                                "display": "General"
                            }
                        ],
                        "text": "General"
                    }
                ],
                "reasonCode": [
                    {
                        "coding": [
                            {
                                "system": "http://snomed.info/sct",
                                "code": "196781001",
                                "display": "Acute appendicitis with peritonitis (disorder)"
```

```
                    }
                ],
                "text": "Acute appendicitis with peritonitis (disorder)"
            }
        ],
        "description": "APPENDECTOMY OPEN [44950 (CPT®)] with Hanson, Steven in MMD East OR at  7:00 AM on
2/9/2022",
        "start": "2022-02-08T20:00:00Z",
        "end": "2022-02-08T22:00:00Z",
        "minutesDuration": 120,
        "created": "2022-02-02",
        "comment": "Schedule at the first case of the day.\\r\\n",
        "patientInstruction": "Patient should not eat or drink after midnight prior to day of surgery.\r\nPlease
 arrange travel plans to/from the hospital.  Do not forget to bring\r\nany medications you are currently taking.
Please leave any valuables at\r\nhome or with someone for safekeeping.\r\n",
        "participant": [
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PART",
                                "display": "Participation"
                            }
                        ],
                        "text": "PART"
                    }
                ],
                "actor": {
                    "reference": "Patient/eUzI-RPN31f14l7ONMTFzaKlVlo9oMKFk3sePNLooHBA3",
                    "display": "Bennett, Edgar"

                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:00:00Z",
                    "end": "2022-02-08T22:00:00Z"
                }
            },
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PPRF",
                                "display": "primary performer"
                            }
                        ],
                        "text": "PPRF"
                    }
                ],
                "actor": {
                    "reference": "Practitioner/eNb.8WB1163dZHCD5WSpXGQ3",
                    "display": "Steven Hanson"
                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:15:00Z",
                    "end": "2022-02-08T21:50:00Z"
                }
            },
            {
                "type": [
                    {
                        "coding": [
                            {
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                                "code": "PART",
                                "display": "Participation"
                            }
                        ],
                        "text": "PART"
                    }
                ],
                "actor": {
                    "reference": "Location/eNvgllF6SIj07NXjl8ww8nw3",
                    "display": "Mmd East Or"
                },
                "required": "required",
                "status": "accepted",
                "period": {
                    "start": "2022-02-08T20:00:00Z",
                    "end": "2022-02-08T22:00:00Z"
                }
            },
            {
```

```
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/e5QKlsvlrLviW4IcugVAxxA3",
            "display": "Circulator Anderson"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/eSLiMCyhfpHC-4zQUgTR9UQ3",
            "display": "Scrub Kennedy"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "SPRF",
                        "display": "secondary performer"
                    }
                ],
                "text": "SPRF"
            }
        ],
        "actor": {
            "reference": "Practitioner/eUAxZ5XXMnRkePszqD0W5xA3",
            "display": "Anesthesiologist Anesthesia, MD"
        },
        "required": "required",
        "status": "accepted",
        "period": {
            "start": "2022-02-08T20:00:00Z",
            "end": "2022-02-08T22:00:00Z"
        }
    },
    {
        "type": [
            {
                "coding": [
                    {
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ParticipationType",
                        "code": "PART",
                        "display": "Participation"
                    }
                ],
                "text": "PART"
            }
        ],
        "actor": {
            "reference": "Location/elxd-pgW0IoEzZEZd0Ifl2Q3",
            "display": "MMD EAST OR 1"
        },
        "required": "required",
```

```
                        "status": "accepted",
                        "period": {
                            "start": "2022-02-08T20:00:00Z",
                            "end": "2022-02-08T22:00:00Z"
                        }
                    }
                ],
                "requestedPeriod": [
                    {
                        "start": "2022-02-03T21:30:00Z",
                        "end": "2022-02-03T23:30:00Z"
                    }
                ]
            },
            "search": {
                "mode": "match"
            }
        }
    ]
}
```

30.2 Patient Workflow Scheduling: STU3

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/index.html

* Appointment.$book (STU3)
* Appointment.$find (STU3)
* Appointment.Read (Appointments) (STU3)
* Appointment.Search (Appointments) (STU3)
* Schedule.Read (STU3)
* Slot.Read (STU3)

30.2.1 Appointment.$book (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/839-Appointment-book-STU3.html
Epic API ID | 839
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
API group | Appointment Group
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | POST
URL template | /api/FHIR/STU3/Appointment/$book
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=839

Description

Note: You must first use the Appointment.$find (STU3) operation before using Appointment.$book (STU3). See the note in Appointment.$find (STU3) regarding pre-coordination with the healthcare organization.

The FHIR Appointment.$book operation schedules an appointment that was returned by the $find operation. It will take a patient ID, appointment FHIR ID from the result selected from the $find results, and optionally a free-text note to store with the appointment.

A typical workflow would involve an end user, who is intending to book an appointment for patient, entering in some basic criteria, such as when the appointment would occur, what specialty or visit type is desired, and optional information about the patient, such as date of birth, sex, or any provider preferences the patient has. The user app would then collect those criteria and submit them using the $find operation. The scheduling system would use organization-defined rules to identify potential appointments slots and returns those for the user to review. The user would review the possible options, and in collaboration with the patient, select the best option. For new patients, the app would perform a Patient.Create interaction. The app would then book the appointment using the $book operation.

$book returns the appointment resource.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Parameters | Parameters
Parameters.parameter | BackboneElement | | Yes
Parameters.parameter.extension (also: Patient) | Extension | | Yes
Parameters.parameter.resource | ResourceContainer | | | | A single resource returned as part of this bundle.

Parameters.parameter.resource.Appointment | Appointment | | | An instance of the STU3 Appointment resource.
Parameters.parameter.resource.Appointment.slot | Reference | | Yes | | Reference to another resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | | Yes | | A single entry in the resource bundle.
Bundle.entry.resource | ResourceContainer | | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Appointment | Appointment | | | | An instance of the STU3 Appointment resource.
Bundle.entry.resource.Appointment.end | instant | | | | The end time.
Bundle.entry.resource.Appointment.id | id
Bundle.entry.resource.Appointment.identifier | Identifier | | Yes | | The CSN.
Bundle.entry.resource.Appointment.minutesDuration | positiveInt | | | | The duration (in minutes).
Bundle.entry.resource.Appointment.participant | BackboneElement | | Yes
Bundle.entry.resource.Appointment.participant.actor | Reference | | | | The FHIR ID reference and display name.
Bundle.entry.resource.Appointment.serviceType | CodeableConcept | | Yes | | Visit type and display name.
Bundle.entry.resource.Appointment.start | instant | | | | The start time.
Bundle.entry.resource.Appointment.status | code | | | | Always "booked."

Sample request

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "patient",
      "valueIdentifier": {
        "value": "efvHwbc1k1CQ9XjM1zvvefQ3"
      }
    },
    {
      "name": "appointment",
      "valueIdentifier": {
        "value": "ezu6MfS.FpOXrHAn1eJHczv4LlH.fMIwtdkA8rsm-Yfu96eUh91EBd0UN9BZx7kbB3"
      }
    },
    {
      "name": "appointmentNote",
      "valueString": "Note text containing info related to the appointment."
    }
  ]
}
```

Sample response

```
{
  "resourceType": "Appointment",
  "id": "eU32HphPn1VaOoKuco8Q56SVsfdrB3fsVeoLAjhpvLY43",
  "identifier": [
    {
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
      "value": "10158405185"
    }
  ],
  "status": "booked",
  "serviceType": [
    {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
          "code": "225",
          "display": "Physical Therapy Visit"
        }
      ]
    }
  ],
  "start": "2018-08-04T00:00:00Z",
  "end": "2018-08-04T00:30:00Z",
  "minutesDuration": 30,
  "participant": [
    {
      "actor": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Patient/efvHwbc1k1CQ9XjM1zvvefQ3",
        "display": "CDS,Fhir"
      },
      "status": "accepted"
    },
    {
      "actor": {
        "reference":
    "https://hostname/instance/api/FHIR/STU3/Practitioner/edNKJE8VjTF.I8BInZ7jSn5nAYzw4T3TaxTbsCoM.lgY3",
        "display": "Jones, Alex, MD"
      },
```

```
      "status": "accepted"
    },
    {
      "actor": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Location/e6mKSqgx8ytPcbu-A4F0TVA3",
        "display": "INITIAL DEPARTMENT"
      },
      "status": "accepted"
    }
  ]
}
```


30.2.2 Appointment.$find (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/840-Appointment-find-STU3.html
Epic API ID | 840
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
API group | Appointment Group
USCDI | No
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | POST
URL template | /api/FHIR/STU3/Appointment/$find
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=840

Description

Note: The $find operation requires pre-coordination with the healthcare organization. The organization must build out specific rules in the Epic Cadence scheduling system that accept the provided inputs and determine which providers and slots will be returned. The request elements described here are the list of *potential* elements that might be used by an organization's scheduling rules. Some organizations may ignore or require any of these properties based on their organizational rules.

The FHIR Appointment.$find operation finds a list of potential appointments slots. For each $find operation run, there will not be overlapping appointment slots returned. For example, you will only receive one appointment slot for 10:00AM-10:30AM. The Appointment IDs in the returned list can be used by the Appointment.$book operation to schedule that appointment.

A typical workflow would involve an end user, who is intending to book an appointment for patient, entering in some basic criteria, such as when the appointment would occur, what specialty or visit type is desired, the patient's time zone for the appointment, and optional information about the patient, such as date of birth, sex, or any provider preferences the patient has. The user app would then collect those criteria and submit them using the $find operation. The scheduling system would use organization-defined rules to identify potential appointments slots, and returns those for the user to review. The user would review the possible options, and in collaboration with the patient, select the best option. For new patients, the app would perform a Patient.Create interaction. The app would then book the appointment using the $book operation.

Remarks

* Appointment.$find does not return existing appointments for patients (past or future).

* The proposed appointment slots returned are only held temporarily.

* The response to $find is paged. By default, only the first 10 results are returned. If those results do not work for the patient, more results can be retrieved by following the "next" pointer to get the next page of results.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
Parameters (STU3) | Parameters
Parameters (STU3).parameter (endTime) (also: parameter (endTime)) | BackboneElement | | Yes
Parameters (STU3).parameter (endTime).name | string | | | 1..1 | Set to endTime.
Parameters (STU3).parameter (endTime).valueDateTime | dateTime | | | 1..1 | Set to the end dateTime of appointment slots to search for.
Parameters (STU3).parameter (indications) (also: parameter (indications)) | BackboneElement | | Yes
Parameters (STU3).parameter (indications).name | string | | | 1..1 | Set to indications.
Parameters (STU3).parameter (indications).valueCodeableConcept | CodeableConcept | | | 1..1 | Diagnosis code indications. This can be a community diagnosis code and/or a SNOMED code. If multiple codes are received, the community diagnosis code takes precedence.
Parameters (STU3).parameter (location-reference) (also: parameter (location-reference)) | BackboneElement | | Yes
Parameters (STU3).parameter (location-reference).name | string | | | 1..1 | Set to location-reference.
Parameters (STU3).parameter (location-reference).valueReference | Reference | | | 1..1 | Reference to another resource.
Parameters (STU3).parameter (location-reference).valueReference.reference | String | | | 1..1 | Reference to the Location resource for where to search for appointments.
Parameters (STU3).parameter (patient) (also: parameter (patient)) | BackboneElement | | Yes
Parameters (STU3).parameter (patient).name | string | | | 1..1 | Set to patient.

Parameters (STU3).parameter (patient).resource | ResourceContainer | | | 1..1 | Contains the Patient FHIR resource. Specify resource/resourceType = Patient

Parameters (STU3).parameter (patient).resource.Patient | Resource | | | 1..1 | The Patient FHIR resource with demographics used to match to a patient record.

Parameters (STU3).parameter (patient).resource.Patient.address (home) (also: address (home)) | Address | | Yes | 0..1 | The patient's home address information.

Parameters (STU3).parameter (patient).resource.Patient.address (home).city | String | | | 0..1 | The city that the patient lives in.

Parameters (STU3).parameter (patient).resource.Patient.address (home).country | String | | | 0..1 | The country the patient lives in.

Parameters (STU3).parameter (patient).resource.Patient.address (home).district | String | | | 0..1 | The patient's county of residence.

Parameters (STU3).parameter (patient).resource.Patient.address (home).line | String | | Yes | 0..* | The patient's street address. You can enter multiple lines of address.

Parameters (STU3).parameter (patient).resource.Patient.address (home).postalCode | String | | | 0..1 | The patient's ZIP code in one of the following formats: nnnnn nnnnn-nnnn

Parameters (STU3).parameter (patient).resource.Patient.address (home).state | String | | | 0..1 | The state where the patient lives. Enter the two-letter postal abbreviation for the state.

Parameters (STU3).parameter (patient).resource.Patient.address (home).use | String | | | 0..1 | Specifies the type of address. The permanent address uses the code home.

Parameters (STU3).parameter (patient).resource.Patient.address (temp) (also: address (temp)) | Address | | Yes | 0..1 | Patient's temporary address information.

Parameters (STU3).parameter (patient).resource.Patient.address (temp).country | String | | | 0..1 | The country the patient lives in.

Parameters (STU3).parameter (patient).resource.Patient.address (temp).state | String | | | 0..1 | The state where the patient lives. Enter the two-letter postal abbreviation for the state.

Parameters (STU3).parameter (patient).resource.Patient.address (temp).use | String | | | 1..1 | Specifies the type of address. The temporary address uses the code temp.

Parameters (STU3).parameter (patient).resource.Patient.birthDate | date | | | 0..1 | The patient's date of birth in the format YYYY-MM-DD.

Parameters (STU3).parameter (patient).resource.Patient.extension (timezone) (also: extension (timezone)) | Extension | | Yes | 0..1 | Patient time zone extension.

Parameters (STU3).parameter (patient).resource.Patient.extension (timezone).url | uri | | | 1..1 | Time zone URL. Must be set to http://hl7.org/fhir/StructureDefinition/timezone

Parameters (STU3).parameter (patient).resource.Patient.extension (timezone).valueCode | code | | | 1..1 | Time zone title.

Parameters (STU3).parameter (patient).resource.Patient.gender | code | | | 0..1 | The patient's legal sex.

Parameters (STU3).parameter (patient).resource.Patient.identifier | Identifier | | Yes | 0..* | The patient's identifier. To look up a patient using only the identifier without any demographics information, refer to Patient.Search (R4). Epic MPI IDs (for example, the Patient's MRN) and national IDs (for example, SSN) are supported in this field. Historical MPI IDs are supported with no special formatting requirements. If an MPI ID is provided, Epic uses the identifier to find the patient instead of the provided demographics. The matching patient is then verified using the provided demographics. If a national ID is provided, Epic performs a match on overall demographics, including the national ID, instead of a lookup on only the identifier. If multiple identifiers are provided, they must all be on the same patient.

Parameters (STU3).parameter (patient).resource.Patient.identifier.system | String | | | 1..1 | The identifier's system value. Valid system values include: MPI ID OID. This system value is highly configurable at each Epic organization. National identifier OIDs including: US Social Security Number: 2.16.840.1.113883.4.1 UK NHS ID: 2.16.840.1.113883.2.1.4.1 Dutch BSN: 2.16.840.1.113883.2.4.6.3 or http://fhir.nl/fhir/NamingSystem/bsn Finnish National ID: 1.2.246.21

Parameters (STU3).parameter (patient).resource.Patient.identifier.value | String | | | 1..1 | The identifier.

Parameters (STU3).parameter (patient).resource.Patient.name | HumanName | | Yes | 0..1 | The patient's name.

Parameters (STU3).parameter (patient).resource.Patient.name.family | String | | | 0..1 | The patient's family name.

Parameters (STU3).parameter (patient).resource.Patient.name.given | String | | Yes | 0..* | The patient's given name. Can include first and middle names.

Parameters (STU3).parameter (patient).resource.Patient.name.use | String | | | 0..1 | Specifies the type of name. The following use types are supported: usual - used for normal use official - formal name

Parameters (STU3).parameter (patient).resource.Patient.telecom | ContactPoint | | Yes | 0..* | Telephone numbers and email addresses for the patient, along with their use (home or work for example).

Parameters (STU3).parameter (patient).resource.Patient.telecom.system | String | | | 1..1 | The type of telecom. Potential values are: email phone

Parameters (STU3).parameter (patient).resource.Patient.telecom.use | String | | | 0..1 | If the telecom entry is a phone number, possible values are: home mobile work

Parameters (STU3).parameter (patient).resource.Patient.telecom.value | String | | | 1..1 | Value of the telecom method. Can be an email address or phone number.

Parameters (STU3).parameter (referrals) (also: parameter (referrals)) | BackboneElement | | Yes

Parameters (STU3).parameter (referrals).name | string | | | 1..1 | Set to referrals.

Parameters (STU3).parameter (referrals).valueCodeableConcept | CodeableConcept | | | 1..1 | Code indicating the referral ID. The referral is used to determine the correct visit type and providers.

Parameters (STU3).parameter (service-type) (also: parameter (service-type)) | BackboneElement | | Yes | 1..1

Parameters (STU3).parameter (service-type).name | string | | | 1..1 | Set to service-type.

Parameters (STU3).parameter (service-type).valueCodeableConcept | CodeableConcept | | | 1..1 | Coded value for the visit type.

Parameters (STU3).parameter (specialties) (also: parameter (specialties)) | BackboneElement | | Yes

Parameters (STU3).parameter (specialties).name | string | | | 1..1 | Set to specialties.

Parameters (STU3).parameter (specialties).valueCodeableConcept | CodeableConcept | | | 1..1 | Department specialty code.

Parameters (STU3).parameter (startTime) (also: parameter (startTime)) | BackboneElement | | Yes

Parameters (STU3).parameter (startTime).name | string | | | 1..1 | Set to startTime.

Parameters (STU3).parameter (startTime).valueDateTime | dateTime | | | 1..1 | Set to the start dateTime of appointment slots to search for.

Parameters (STU3).parameter (time-of-day) (also: parameter (time-of-day)) | BackboneElement | | Yes

Parameters (STU3).parameter (time-of-day).name | string | | | 1..1 | Set to time-of-day.

Parameters (STU3).parameter (time-of-day).valueTiming | Timing | | | 1..1 | The preferred time and duration for the appointments.

Parameters (STU3).parameter (time-of-day).valueTiming.repeat | BackboneElement | | | 1..1 | Element indicating an event that occurs multiple times.

Parameters (STU3).parameter (time-of-day).valueTiming.repeat.dayOfWeek | code | | Yes | 0..* | Days of the week to search for appointments.

Parameters (STU3).parameter (time-of-day).valueTiming.repeat.duration | decimal | | | 0..1 | The duration of the appointment.

Parameters (STU3).parameter (time-of-day).valueTiming.repeat.durationUnit | code | | | 0..1 | Duration units. Although a value of 'min' makes the most sense, all of the following values are supported: s - seconds min - minutes h - hours d - days

Parameters (STU3).parameter (time-of-day).valueTiming.repeat.timeOfDay | time | | Yes | 0..1 | Specific time of the day for the appointment to take place.

Response elements

Name | Type | Optional | Array | Cardinality | Description

Appointment (STU3) | Appointment | | | | An instance of the STU3 Appointment resource.

Appointment (STU3).appointmentType | CodeableConcept | | | 0..1 | The type of appointment.

Appointment (STU3).comment | string | | | 0..1 | The appointment notes (limited to 500 characters).

Appointment (STU3).contained | ResourceContainer | | Yes | 0..* | Contained resources.

Appointment (STU3).contained.Slot | Resource | | | 1..1 | The Slot(s) that this appointment is filling. Slots are sent as contained resources.

Appointment (STU3).contained.Slot.end | instant | | | 1..1 | Appointment slot end time.

Appointment (STU3).contained.Slot.id | id | | | 1..1 | Slot FHIR ID.

Appointment (STU3).contained.Slot.schedule | Reference | | | 0..1 | References to the Schedule resource.

Appointment (STU3).contained.Slot.schedule.reference | String | | | 0..1 | Schedule FHIR ID.

Appointment (STU3).contained.Slot.start | instant | | | 1..1 | Appointment slot start time.

Appointment (STU3).contained.Slot.status | code | | | 1..1 | Appointment slot status.

Appointment (STU3).created | dateTime | | | 0..1 | The date when the appointment was created.

Appointment (STU3).end | instant | | | 0..1 | The appointment end time.

Appointment (STU3).extension (timezone) (also: extension (timezone)) | Extension | | Yes | 0..1 | Appointment department time zone. URL: http://hl7.org/fhir/StructureDefinition/timezone

Appointment (STU3).extension (timezone).valueCode | code | | | 1..1 | Timezone code.

Appointment (STU3).id | id | | | 1..1 | The Appointment FHIR ID. The Appointment resource IDs returned here can only be used with the $book FHIR API. You cannot perform a FHIR Read on these potential appointments.

Appointment (STU3).identifier | Identifier | | Yes | 1..* | The appointment CSN (Epic Contact Serial Number).

Appointment (STU3).meta | Meta | | | 0..1 | The document metadata.

Appointment (STU3).meta.profile | String | | Yes | 0..1 | Only returned for organizations in the Netherlands. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/eAfspraak-Appointment

Appointment (STU3).minutesDuration | positiveInt | | | 0..1 | The estimated length of the appointment (in minutes).

Appointment (STU3).participant | BackboneElement | | Yes | 1..* | Patient, providers, locations, and scheduled resources (such as MRI machines) that are associated with the appointment.

Appointment (STU3).participant.actor | Reference | | | | FHIR ID reference and display name.

Appointment (STU3).participant.required | code | | | 1..1 | Always set to "required".

Appointment (STU3).participant.status | code | | | 1..1 | The appointment status.

Appointment (STU3).reason | CodeableConcept | | Yes | 0..* | The reason for the appointment.

Appointment (STU3).serviceCategory | CodeableConcept | | | | Service category.

Appointment (STU3).serviceCategory.coding | Coding | | Yes | 1..1 | Service category code. Always set to "appointment".

Appointment (STU3).serviceCategory.coding.code | String | | | 1..1 | Always set to "appointment".

Appointment (STU3).serviceCategory.coding.display | String | | | 1..1 | Always set to "Appointment".

Appointment (STU3).serviceCategory.coding.system | String | | | 1..1 | Always set to "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category".

Appointment (STU3).serviceCategory.text | String | | | 1..1 | Always set to "appointment".

Appointment (STU3).serviceType | CodeableConcept | | Yes | 0..* | The visit type.

Appointment (STU3).slot | Reference | | Yes | 0..* | The Slot(s) that this appointment is filling. Slots are sent as contained resources.

Appointment (STU3).slot.display | String | | | 0..1 | Slot provider's name.

Appointment (STU3).slot.reference | String | | | 1..1 | The Slot FHIR ID.

Appointment (STU3).specialty | CodeableConcept | | Yes | 0..* | The specialty for the appointment.

Appointment (STU3).start | instant | | | 0..1 | The appointment start time.

Appointment (STU3).status | code | | | 1..1 | The appointment status.

Appointment (STU3).status.extension (code-specification) (also: extension (code-specification)) | Extension | | | 0..1 | This extension is returned only for organizations in the Netherlands. The order status for the order associated with the appointment. URL: http://nictiz.nl/fhir/StructureDefinition/code-specification

Appointment (STU3).status.extension (code-specification).valueCodeableConcept | CodeableConcept | | | 1..1 | Additional coded status values.

Sample request

```
{"resourceType":"Parameters","parameter":[
    {"name":"patient",
    "resource":{
    "resourceType":"Patient",
    "extension":[
    {"valueCode":"M","url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex"},
    {"extension":[
    {"valueCoding":{"system":"http://hl7.org/fhir/us/core/ValueSet/omb-race-
    category","code":"2106-3","display":"White"},"url":"ombCategory"},
    {"valueString":"White","url":"text"}],
    "url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"},
    {"extension":[
    {"valueCoding":{"system":"http://hl7.org/fhir/us/core/ValueSet/omb-ethnicity-
    category","code":"UNK","display":"Unknown"},
    "url":"ombCategory"},
    {"valueString":"Unknown","url":"text"}],
    "url":"http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"}],
    "identifier":[
    {"use":"usual",
```

```
    "type":{"text":"EPIC"},
    "system":"urn:oid:1.2.840.114350.1.1",
    "value":"E3423"},
    {"use":"usual",
    "type":{"text":"MRN"},
    "system":"urn:oid:1.2.840.114350.1.13.0.1.7.5.737384.14",
    "value":"203177"}],
    "active":"true",
    "name":[{"use":"usual",
    "text":"Correct Professional Billing",
    "family":"Professional Billing",
    "given":["Correct"]}],
    "telecom":[
    {"system":"phone",
    "value":"608-271-9000",
    "use":"home"},
    {"system":"phone",
    "value":"608-271-9000","use":"work"}],
    "gender":"male",
    "birthDate":"1983-06-08",
    "deceasedBoolean":false,
    "address":[
    {"use":"home",
    "line":["1979 Milky Way"],
    "city":"VERONA",
    "district":"DANE",
    "state":"WI",
    "postalCode":"53593",
    "country":"US"}],
    "maritalStatus":{"text":"Single"},
    "communication":[
    {"language":{"coding":[
    {"system":"http://hl7.org/fhir/ValueSet/languages",
    "code":"en",
    "display":"English"}],
    "text":"English"},
    "preferred":"true"}],
    "generalPractitioner":[
    {"reference":"https://apporchard.epic.com/interconnect-aocurprd-
    oauth/api/FHIR/STU3/Practitioner/eM5CWtq15N0WJeuCet5bJlQ3",
    "display":"Physician Family Medicine, MD"}],
    "managingOrganization":{"reference":"https://apporchard.epic.com/interconnect-aocurprd-
    oauth/api/FHIR/STU3/Organization/enRyWnSP963FYDpoks4NHOA3",
    "display":"Epic Hospital System"}}},
    {"name":"startTime",
    "valueDateTime":"2024-05-01T13:00:00Z"},
    {"name":"endTime",
    "valueDateTime":"2028-05-05T22:00:00Z"},
    {"name":"serviceType",
    "valueCodeableConcept":{"coding":[{"system":"urn:oid:1.2.840.114350.1.13.0.1.7.3.808267.11",
    "code":"95014",
    "display":"Office Visit"}]}},
    {"name":"indications",
    "valueCodeableConcept":{"coding":[
    {"system":"urn:oid:2.16.840.1.113883.6.96",
    "code":"46866001",
    "display":"Fracture of lower limb (disorder)"},
    {"system":"urn:oid:2.16.840.1.113883.6.90",
    "code":"S82.90XA",
    "display":"Broken leg"},
    {"system":"urn:oid:1.2.840.114350.1.13.861.1.7.2.696871",
    "code":"121346631",
    "display":"Broken leg"}],
    "text":"Broken leg"}},
    {"name":"location-reference",
    "valueReference":{"reference":"https://apporchard.epic.com/interconnect-aocurprd-
    oauth/api/FHIR/STU3/Location/e4W4rmGe9QzuGm2Dy4NBqVc0KDe6yGld6HW95UuN-Qd03"
    }}]
```


Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 5,
  "link": [
    {
      "relation": "next",
      "url":
    "https://hostname/instance/api/FHIR/STU3/Appointment/$find?_count=5&sessionID=14-9BB80932942E11E8B7AE460231621F93"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url":
    "https://hostname/instance/api/FHIR/STU3/Appointment/eogfzNPjsx55fSvPUEEThHfbtONejGmNXQKllRiY6kPhRV9hXAHVL6OcPAOwree
```

```
jz3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/eogfzNPjsx55fSvPUEEThHfbtONejGmNXQKllRiY6kPhRV9hXA
HVL6OcPAOwreejz3",
  "resource": {
    "resourceType": "Appointment",
    "id": "eogfzNPjsx55fSvPUEEThHfbtONejGmNXQKllRiY6kPhRV9hXAHVL6OcPAOwreejz3",
    "contained": [
      {
        "resourceType": "Slot",
        "id": "enA9Zi4LG5esVkLtiqg-ZtQGK6d6mwljGVTZduOQaXmjh8qD10m4ObCmPVRzSrgcG.l8dVBRIuAlPx9BBFuOBFA3",
        "schedule": {
          "reference":
"https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"
        },
        "status": "busy-tentative",
        "start": "2018-07-30T19:30:00Z",
        "end": "2018-07-30T20:30:00Z"
      }
    ],
    "extension": [
      {
        "valueCode": "America/Chicago",
        "url": "http://hl7.org/fhir/StructureDefinition/timezone"
      }
    ],
    "status": "proposed",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "10770",
            "display": "Office Visit"
          }
        ]
      }
    ],
    "start": "2018-07-30T19:30:00Z",
    "end": "2018-07-30T20:30:00Z",
    "minutesDuration": 60,
    "slot": [
      {
        "reference": "#enA9Zi4LG5esVkLtiqg-ZtQGK6d6mwljGVTZduOQaXmjh8qD10m4ObCmPVRzSrgcG.l8dVBRIuAlPx9BBFuOBFA3",
        "display": "Physician Family Medicine"
      }
    ],
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eJr2Age0AyrEID2qZgA4C4F-d3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/egnGs9HJuacSWEJaiACz04Q3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/eULujY-VWnFz-tbLqZ39RjA3",
          "display": "EMC Family Practice"
        },
        "status": "tentative"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Appointment/eZr0n3n3a0y3BMAnKCGhY7khytNzHgPyEZS0FIdpCqEfzhwy7QEmQWGvSrf92WP
rB3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/eZr0n3n3a0y3BMAnKCGhY7khytNzHgPyEZS0FIdpCqEfzhwy7Q
EmQWGvSrf92WPrB3",
  "resource": {
    "resourceType": "Appointment",
```

```
    "id": "eZr0n3n3a0y3BMAnKCGhY7khytNzHgPyEZS0FIdpCqEfzhwy7QEmQWGvSrf92WPrB3",
    "contained": [
      {
        "resourceType": "Slot",
        "id": "enA9Zi4LG5esVkLtiqg-ZtReOlaLWMjuigGI-cGQAPIWtbiXbYlobqBrBiEdb09aRpj4RY2UYuF3UancZesKzag3",
        "schedule": {
          "reference":
"https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"
        },
        "status": "busy-tentative",
        "start": "2018-07-30T19:45:00Z",
        "end": "2018-07-30T20:45:00Z"
      }
    ],
    "extension": [
      {
        "valueCode": "America/Chicago",
        "url": "http://hl7.org/fhir/StructureDefinition/timezone"
      }
    ],
    "status": "proposed",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "10770",
            "display": "Office Visit"
          }
        ]
      }
    ],
    "start": "2018-07-30T19:45:00Z",
    "end": "2018-07-30T20:45:00Z",
    "minutesDuration": 60,
    "slot": [
      {
        "reference": "#enA9Zi4LG5esVkLtiqg-ZtReOlaLWMjuigGI-cGQAPIWtbiXbYlobqBrBiEdb09aRpj4RY2UYuF3UancZesKzag3",
        "display": "Physician Family Medicine"
      }
    ],
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eJr2Age0AyrEID2qZgA4C4F-d3",
          "display": "Family Medicine Physician"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/egnGs9HJuacSWEJaiACz04Q3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/eULujY-VWnFz-tbLqZ39RjA3",
          "display": "EMC Family Medicine"
        },
        "status": "tentative"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Appointment/e90p1pXMU83z9wukoqd59V1hEPIKmvDfdBbamaB8cv1qubYrmSZzXXG3Jk-
I34D7S3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/e90p1pXMU83z9wukoqd59V1hEPIKmvDfdBbamaB8cv1qubYrmS
ZzXXG3Jk-I34D7S3",
  "resource": {
    "resourceType": "Appointment",
    "id": "e90p1pXMU83z9wukoqd59V1hEPIKmvDfdBbamaB8cv1qubYrmSZzXXG3Jk-I34D7S3",
    "contained": [
      {
        "resourceType": "Slot",
        "id": "enA9Zi4LG5esVkLtiqg-ZtWACmGICfDCt0qxqjUJUC6JxvgGWYR57T3mN754htF6x35rPv-j1GsldRpoSE4sSVA3",
        "schedule": {
          "reference":
```

```
"https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"
        },
        "status": "busy-tentative",
        "start": "2018-07-30T20:00:00Z",
        "end": "2018-07-30T21:00:00Z"
      }
    ],
    "extension": [
      {
        "valueCode": "America/Chicago",
        "url": "http://hl7.org/fhir/StructureDefinition/timezone"
      }
    ],
    "status": "proposed",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "10770",
            "display": "Office Visit"
          }
        ]
      }
    ],
    "start": "2018-07-30T20:00:00Z",
    "end": "2018-07-30T21:00:00Z",
    "minutesDuration": 60,
    "slot": [
      {
        "reference": "#enA9Zi4LG5esVkLtiqg-ZtWACmGICfDCt0qxqjUJUC6JxvgGWYR57T3mN754htF6x35rPv-j1GsldRpoSE4sSVA3",
        "display": "Physician Family Medicine"
      }
    ],
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eJr2Age0AyrEID2qZgA4C4F-d3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/egnGs9HJuacSWEJaiACz04Q3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/eULujY-VWnFz-tbLqZ39RjA3",
          "display": "EMC Family Medicine"
        },
        "status": "tentative"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Appointment/eOUMf6ePfyHoParQS5LGX98Q1pl.z3QrTZZE0MvjmHNUJdzLGJ6etvU2.OEkb6l
6L3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/eOUMf6ePfyHoParQS5LGX98Q1pl.z3QrTZZE0MvjmHNUJdzLGJ
6etvU2.OEkb6l6L3",
  "resource": {
    "resourceType": "Appointment",
    "id": "eOUMf6ePfyHoParQS5LGX98Q1pl.z3QrTZZE0MvjmHNUJdzLGJ6etvU2.OEkb6l6L3",
    "contained": [
      {
        "resourceType": "Slot",
        "id": "enA9Zi4LG5esVkLtiqg-ZtYTh7WHzSH-wSvLwjypmfwdljWaLLZjgO0vNQvzgTSmfNy-PLQP-RlL6t40RhWB1OQ3",
        "schedule": {
          "reference":
"https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"
        },
        "status": "busy-tentative",
        "start": "2018-07-30T20:15:00Z",
        "end": "2018-07-30T21:15:00Z"
      }
    ],
```

```
    "extension": [
      {
        "valueCode": "America/Chicago",
        "url": "http://hl7.org/fhir/StructureDefinition/timezone"
      }
    ],
    "status": "proposed",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "10770",
            "display": "Office Visit"
          }
        ]
      }
    ],
    "start": "2018-07-30T20:15:00Z",
    "end": "2018-07-30T21:15:00Z",
    "minutesDuration": 60,
    "slot": [
      {
        "reference": "#enA9Zi4LG5esVkLtiqg-ZtYTh7WHzSH-wSvLwjypmfwdljWaLLZjgO0vNQvzgTSmfNy-PLQP-RlL6t40RhWB1OQ3",
        "display": "Physician Family Medicine"
      }
    ],
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eJr2Age0AyrEID2qZgA4C4F-d3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/egnGs9HJuacSWEJaiACz04Q3",
          "display": "Physician Family Medicine"
        },
        "status": "tentative"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/eULujY-VWnFz-tbLqZ39RjA3",
          "display": "EMC Family Medicine"
        },
        "status": "tentative"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url":
"https://hostname/instance/api/FHIR/STU3/Appointment/eugZHA48Y6O.03bhMkOHpnvxt1L888cBw2m3g2JSkNn2QLEC3abrEmN-
EFyMlSMlE3"
    }
  ],
  "fullUrl":
"https://hostname/instance/api/FHIR/STU3/Appointment/eugZHA48Y6O.03bhMkOHpnvxt1L888cBw2m3g2JSkNn2QLEC3abrEmN-
EFyMlSMlE3",
  "resource": {
    "resourceType": "Appointment",
    "id": "eugZHA48Y6O.03bhMkOHpnvxt1L888cBw2m3g2JSkNn2QLEC3abrEmN-EFyMlSMlE3",
    "contained": [
      {
        "resourceType": "Slot",
        "id": "enA9Zi4LG5esVkLtiqg-Zte3wlRB0rhTyXCPK3hSC4di1vfe9J5cBiJUTOphoKPlP6Qiec.Ivfxnzr-RZLAqfqg3",
        "schedule": {
          "reference":
"https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"
        },
        "status": "busy-tentative",
        "start": "2018-07-30T20:30:00Z",
        "end": "2018-07-30T21:30:00Z"
      }
    ],
    "extension": [
      {
        "valueCode": "America/Chicago",
        "url": "http://hl7.org/fhir/StructureDefinition/timezone"
      }
    ],
```

```
        "status": "proposed",
        "serviceType": [
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
                "code": "10770",
                "display": "Office Visit"
              }
            ]
          }
        ],
        "start": "2018-07-30T20:30:00Z",
        "end": "2018-07-30T21:30:00Z",
        "minutesDuration": 60,
        "slot": [
          {
            "reference": "#enA9Zi4LG5esVkLtiqg-Zte3wlRB0rhTyXCPK3hSC4di1vfe9J5cBiJUTOphoKPlP6Qiec.Ivfxnzr-RZLAqfqg3",
            "display": "Physician Family Medicine"
          }
        ],
        "participant": [
          {
            "actor": {
              "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eJr2Age0AyrEID2qZgA4C4F-d3",
              "display": "Physician Family Medicine"
            },
            "status": "tentative"
          },
          {
            "actor": {
              "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/egnGs9HJuacSWEJaiACz04Q3",
              "display": "Physician Family Medicine"
            },
            "status": "tentative"
          },
          {
            "actor": {
              "reference": "https://hostname/instance/api/FHIR/STU3/Location/eULujY-VWnFz-tbLqZ39RjA3",
              "display": "EMC Family Medicine"
            },
            "status": "tentative"
          }
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```


Change log

Epic Version | Description | Type
May 2022 | The Participant.actor parameter returns scheduled resources, such as MRI machines. | Non-substantive Change

30.2.3 Appointment.Read (Appointments) (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/466-Appointment-Read-Appointments-STU3.html
Epic API ID | 466
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Appointment/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=466

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for an Appointment resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Appointment | Appointment | False | | | An instance of the STU3 Appointment resource.
Appointment.extension (also: Appointment.Status OrderStatus) | Extension | True | Yes | | The order status for the order associated with the appointment. This extension is returned only for organizations in the Netherlands.
Appointment.appointmentType | CodeableConcept | True | | | The type of appointment.
Appointment.comment | string | True | | | The appointment notes.
Appointment.created | dateTime | True | | | The date when the appointment was created.
Appointment.end | instant | True | | | The appointment end time.
Appointment.identifier | Identifier | True | Yes | | The CSN.
Appointment.meta | Meta | True | | | The document metadata.
Appointment.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/eAfspraak-Appointment
Appointment.minutesDuration | positiveInt | True | | | The estimated length of the appointment.
Appointment.participant | BackboneElement | False | Yes | | Patient, providers, and locations associated with the appointment. Starting in the May 22 version of Epic, scheduled resources, such as MRI machines, are also returned.
Appointment.reason | CodeableConcept | True | Yes | | The reason for the appointment.
Appointment.serviceType | CodeableConcept | True | Yes | | The visit type.
Appointment.slot | Reference | True | Yes | | The slot or slots that this appointment is filling.
Appointment.specialty | CodeableConcept | True | Yes | | The specialty for the appointment.
Appointment.start | instant | True | | | The appointment start time.
Appointment.status | code | False | | | The appointment status.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Appointment/eW2tA04q1RtmXMBG-w4s2PhafwRhG0X0dCPaVJ.4cDh83
```

Sample response


```
{
  "resourceType": "Appointment",
  "id": "ePjxkyjA8gju08Vwqc.iiAFHBGCmkucuk3O15LOr0KFg3",
  "identifier": [
    {
      "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
      "value": "10001659236"
    }
  ],
  "status": "proposed",
  "serviceCategory": {
    "coding": [
      {
        "system": "http://open.epic.com/FHIR/StructureDefinition/appointment-service-category",
        "code": "appointment",
        "display": "Appointment"
      }
    ],
    "text": "appointment"
  },
  "serviceType": [
    {
      "coding": [
        {
          "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
          "code": "579",
          "display": "ABF Office Visit"
        }
      ]
    }
  ],
  "appointmentType": {
    "coding": [
      {
        "system": "http://hl7.org/fhir/v3/ActCode",
        "code": "AMB",
        "display": "Ambulatory"
      }
    ]
  },
  "reason": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "271799000",
          "display": "Head movements abnormal (finding)"
        }
      ],
      "text": "Abnormal head movements"
    }
  ],
  "start": "2021-08-22T13:15:00Z",
  "end": "2021-08-22T13:30:00Z",
```

```
  "minutesDuration": 15,
  "created": "2021-08-20",
  "comment": "Patient instructions are spiffy!\n\nThis will get you to google!\n",
  "participant": [
    {
      "actor": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eFs2zvgmbGfgWFfHliSRYZA3",
        "display": "Gibson,Ken"
      },
      "status": "tentative"
    },
    {
      "actor": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/euc69RmkeUC5UjZOIGu0FiA3",
        "display": "Amanda Fahr"
      },
      "status": "tentative"
    },
    {
      "actor": {
        "reference": "https://hostname/instance/api/FHIR/STU3/Location/e6gRswU5WJtj7msgU7NZiYw3",
        "display": "ABF Family Practice"
      },
      "status": "tentative"
    }
  ]
}
```


Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96
May 2022 | Starting in Epic version May 2022, the OID system structure for CSNs will no longer contain the value of the CSN. Instead, the system will be a static value for all CSNs for a single environment. This lets clients verify a value is unique under a given system. | Breaking change

30.2.4 Appointment.Search (Appointments) (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/10189-Appointment-Search-Appointments-STU3.html
Epic API ID | 10189
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | November 2021
HTTP method | GET
URL template | /api/FHIR/STU3/Appointment?date={date}&identifier={identifier}&patient={patient}&status={status}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10189

Description

Request parameters

Name | Type | Optional | Array | Cardinality | Description
date | String | True | | | Appointment date.
identifier | String | True | | | Appointment identifier (encounter CSN) in the format <code system>|<code>.
patient | String | False | | | Reference to the Patient resource.
status | String | True | | | Appointment status. Supported values include: booked fulfilled cancelled noshow arrived proposed

Response elements

Name | Type | Optional | Array | Cardinality | Description
Bundle | Bundle | False | | | Resource bundle. For more information, refer to the FHIR specification.
Bundle.entry | Entry | False | Yes | | A single entry in the resource bundle.

Bundle.entry.resource | ResourceContainer | False | | | A single resource returned as part of this bundle.
Bundle.entry.resource.Appointment | Appointment | False | | | An instance of the STU3 Appointment resource.
Bundle.entry.resource.Appointment.extension (also: Appointment.status OrderStatus) | Extension | True | Yes | | The order status for the order associated with the appointment. This extension is returned only for organizations in the Netherlands.
Bundle.entry.resource.Appointment.appointmentType | CodeableConcept | True | | | The type of appointment.
Bundle.entry.resource.Appointment.comment | string | True | | | The appointment notes.
Bundle.entry.resource.Appointment.created | dateTime | True | | | The date when the appointment was created.
Bundle.entry.resource.Appointment.end | instant | True | | | The appointment end time.
Bundle.entry.resource.Appointment.id | id | False | | | The appointment FHIR ID.
Bundle.entry.resource.Appointment.identifier | Identifier | False | Yes | | The appointment CSN.
Bundle.entry.resource.Appointment.meta | Meta | True | | | The document metadata.
Bundle.entry.resource.Appointment.meta.profile | String | True | | | Available in Netherlands environments only. This environment shows that this API conforms to the profile connected to the following canonical url: http://nictiz.nl/fhir/StructureDefinition/eAfspraak-Appointment
Bundle.entry.resource.Appointment.minutesDuration | positiveInt | True | | | The estimated length of the appointment.
Bundle.entry.resource.Appointment.participant | BackboneElement | False | Yes | | Patient, providers, and locations associated with the appointment. Starting in the May 22 version of Epic, scheduled resources, such as MRI machines, are also returned.
Bundle.entry.resource.Appointment.reason | CodeableConcept | True | Yes | | The reason for the appointment.
Bundle.entry.resource.Appointment.serviceType | CodeableConcept | True | Yes | | The visit type.
Bundle.entry.resource.Appointment.specialty | CodeableConcept | True | Yes | | The specialty for the appointment.
Bundle.entry.resource.Appointment.start | instant | True | | | The appointment start time.
Bundle.entry.resource.Appointment.status | code | False | | | The appointment status.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Appointment?patient=eFs2zvgmbGfgWFfHliSRYZA3&status=noshow
```

Sample response


```
{
  "resourceType": "Bundle",
  "type": "searchset",
  "total": 4,
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Appointment?patient=eFs2zvgmbGfgWFfHliSRYZA3&status=noshow"
    }
  ],
  "entry": [
    {
      "link": [
        {
          "relation": "self",
          "url": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAJo7tRwZjEdilY-0xzgcL2w3"
        }
      ],
      "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAJo7tRwZjEdilY-0xzgcL2w3",
      "resource": {
        "resourceType": "Appointment",
        "id": "ePjxkyjA8gju08Vwqc.iiAJo7tRwZjEdilY-0xzgcL2w3",
        "identifier": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
            "value": "10001639946"
          }
        ],
        "status": "noshow",
        "serviceType": [
          {
            "coding": [
              {
                "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
                "code": "1667",
                "display": "ABF Dentistry"
              }
            ]
          }
        ],
        "appointmentType": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/v3/ActCode",
              "code": "AMB",
              "display": "Ambulatory"
            }
          ]
        },
        "start": "2021-08-13T16:00:00Z",
        "end": "2021-08-13T16:20:00Z",
        "minutesDuration": 20,
        "created": "2021-08-13",
        "participant": [
          {
            "actor": {
```

```
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eFs2zvgmbGfgWFfHliSRYZA3",
          "display": "Gibson,Ken"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/elUvYgT4WTHTjmMacbM0c5g3",
          "display": "Dr. Lou Smoller"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference":
"https://hostname/instance/api/FHIR/STU3/Location/enn9bCfLfFHzucg4xbI7Jsy7kt7-znccUg26HfBqxQjo3",
          "display": "MyChart Dentistry"
        },
        "status": "declined"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAPwx9VIfkAmBkO0QzHl0ZbQ3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAPwx9VIfkAmBkO0QzHl0ZbQ3",
  "resource": {
    "resourceType": "Appointment",
    "id": "ePjxkyjA8gju08Vwqc.iiAPwx9VIfkAmBkO0QzHl0ZbQ3",
    "identifier": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001644619"
      }
    ],
    "status": "noshow",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "579",
            "display": "ABF Office Visit"
          }
        ]
      }
    ],
    "appointmentType": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.5.4",
          "code": "AMB",
          "display": "Ambulatory"
        }
      ]
    },
    "start": "2021-08-16T18:00:00Z",
    "end": "2021-08-16T18:15:00Z",
    "minutesDuration": 15,
    "created": "2021-08-16",
    "comment": "Patient instructions  \n\nLink to Google\n",
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eFs2zvgmbGfgWFfHliSRYZA3",
          "display": "Gibson,Ken"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/euc69RmkeUC5UjZOIGu0FiA3",
          "display": "Amanda Fahr"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/e6gRswU5WJtj7msgU7NZiYw3",
          "display": "ABF Family Practice"
        },
```

```
        "status": "declined"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAAqNojpT2hDqLoM1pkTF0Rw3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAAqNojpT2hDqLoM1pkTF0Rw3",
  "resource": {
    "resourceType": "Appointment",
    "id": "ePjxkyjA8gju08Vwqc.iiAAqNojpT2hDqLoM1pkTF0Rw3",
    "identifier": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001644621"
      }
    ],
    "status": "noshow",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "577",
            "display": "AUS Office Visit"
          }
        ]
      }
    ],
    "specialty": [
      {
        "coding": [
          {
            "system": "urn:oid:2.16.840.1.113883.6.101",
            "code": "207KA0200X",
            "display": "Allergy Physician"
          }
        ],
        "text": "1"
      }
    ],
    "appointmentType": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.5.4",
          "code": "IMP",
          "display": "Inpatient Encounter"
        }
      ]
    },
    "reason": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.728286",
            "code": "41723"
          }
        ],
        "text": "Head Bleed"
      }
    ],
    "start": "2021-08-16T18:15:00Z",
    "end": "2021-08-16T18:30:00Z",
    "minutesDuration": 15,
    "created": "2021-08-16",
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eFs2zvgmbGfgWFfHliSRYZA3",
          "display": "Gibson,Ken"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/ejTHcJyAyLP9ittoMQVgIFg3",
          "display": "Autumn Schmidt, MD"
        },
        "status": "declined"
      },
      {
```

```
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/e-ihfN9W6NFF9fqHMGzRyBw3",
          "display": "AUS Family Medicine"
        },
        "status": "declined"
      }
    ]
  },
  "search": {
    "mode": "match"
  }
},
{
  "link": [
    {
      "relation": "self",
      "url": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAFHBGCmkucuk3O15LOr0KFg3"
    }
  ],
  "fullUrl": "https://hostname/instance/api/FHIR/STU3/Appointment/ePjxkyjA8gju08Vwqc.iiAFHBGCmkucuk3O15LOr0KFg3",
  "resource": {
    "resourceType": "Appointment",
    "id": "ePjxkyjA8gju08Vwqc.iiAFHBGCmkucuk3O15LOr0KFg3",
    "identifier": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.3.698084.8",
        "value": "10001659236"
      }
    ],
    "status": "noshow",
    "serviceType": [
      {
        "coding": [
          {
            "system": "urn:oid:1.2.840.114350.1.13.861.1.7.2.808267",
            "code": "579",
            "display": "ABF Office Visit"
          }
        ]
      }
    ],
    "appointmentType": {
      "coding": [
        {
          "system": "urn:oid:2.16.840.1.113883.5.4",
          "code": "AMB",
          "display": "Ambulatory"
        }
      ]
    },
    "reason": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "271799000",
            "display": "Head movements abnormal (finding)"
          }
        ],
        "text": "Abnormal head movements"
      }
    ],
    "start": "2021-08-22T13:15:00Z",
    "end": "2021-08-22T13:30:00Z",
    "minutesDuration": 15,
    "created": "2021-08-20",
    "comment": "Patient instructions  \n\nLink to Google\n",
    "participant": [
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Patient/eFs2zvgmbGfgWFfHliSRYZA3",
          "display": "Gibson,Ken"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Practitioner/euc69RmkeUC5UjZOIGu0FiA3",
          "display": "Amanda Fahr"
        },
        "status": "declined"
      },
      {
        "actor": {
          "reference": "https://hostname/instance/api/FHIR/STU3/Location/e6gRswU5WJtj7msgU7NZiYw3",
          "display": "ABF Family Practice"
        },
        "status": "declined"
      }
```

```
        ]
      },
      "search": {
        "mode": "match"
      }
    }
  ]
}
```

Change log

Epic Version | Description | Type
November 2022 | Code systems for the following code sets now return a URL in place of an OID: ActCode, ICD-10-CM, ICD-10-CA, ICD-10-NL, SNOMED System URL OID ActCode STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode 2.16.840.1.113883.5.4 ICD-10-CM http://hl7.org/fhir/sid/icd-10-cm 2.16.840.1.113883.6.90 ICD-10-CA http://hl7.org/fhir/sid/icd-10-ca 2.16.840.1.113883.6.94 ICD-10-NL http://hl7.org/fhir/sid/icd-10-nl 2.16.840.1.113883.6.3.2 SNOMED http://snomed.info/sct 2.16.840.1.113883.6.96 | Breaking change
System | URL | OID
ActCode | STU3: http://hl7.org/fhir/v3/ActCode R4: http://terminology.hl7.org/CodeSystem/v3-ActCode | 2.16.840.1.113883.5.4
ICD-10-CM | http://hl7.org/fhir/sid/icd-10-cm | 2.16.840.1.113883.6.90
ICD-10-CA | http://hl7.org/fhir/sid/icd-10-ca | 2.16.840.1.113883.6.94
ICD-10-NL | http://hl7.org/fhir/sid/icd-10-nl | 2.16.840.1.113883.6.3.2
SNOMED | http://snomed.info/sct | 2.16.840.1.113883.6.96

30.2.5 Schedule.Read (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/500-Schedule-Read-STU3.html
Epic API ID | 500
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
API group | Appointment Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Schedule/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=500

Description

The Schedule resource is the link from a slot to a practitioner and location for an appointment.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a Schedule resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Schedule | Schedule | False | | | An instance of the STU3 Schedule resource.
Schedule.actor | Reference | False | Yes | | Reference to a Practitioner or Location resource.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Schedule/epHlcxmeGNzYSXeg2BJmHLUe.kfCjA5sXU50ptkVX8Vo3
```

Sample response

```
<Schedule xmlns="http://hl7.org/fhir">
  <id value="epHlcxmeGNzYSXeg2BJmHLUe.kfCjA5sXU50ptkVX8Vo3"/>
  <actor>
    <reference value="https://hostname/instance/api/FHIR/STU3/Practitioner/euj3vVq.XS7gwizdNR-HTgg3"/>
    <display value="Provider Ecnw, MD"/>
  </actor>
  <actor>
    <reference value="https://hostname/instance/api/FHIR/STU3/Location/en9.gx8URXBowZGtf5pc0oBBVydCX5LUs5hE4nlE8LlI3"/>
    <display value="ECNW Department"/>
  </actor>
</Schedule>
```

30.2.6 Slot.Read (STU3)

File: site/apis/30-Patient-Workflow-Scheduling/2-STU3/862-Slot-Read-STU3.html
Epic API ID | 862
FHIR version | STU3
Categories | Patient Workflow Scheduling
API type | Industry-Standard
API group | Appointment Group
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | Epic 2018
HTTP method | GET
URL template | /api/FHIR/STU3/Slot/{ID}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=862

Description

The FHIR Slot resource is accessible only from the Appointment $find resource. This resource provides time-slots that can be booked using an appointment. They do not provide any information about appointments that are available, just the time, and optionally what the time can be used for.

Request parameters

Name | Type | Optional | Array | Cardinality | Description
ID | String | False | | | FHIR identifier for a Slot resource.

Response elements

Name | Type | Optional | Array | Cardinality | Description
Slot | Resource | False | | | An instance of the STU3 Slot resource.
Slot.end | instant | False | | | The end time.
Slot.schedule | Reference | False | | | This is how we show the associated Patient/Practitioner.
Slot.start | instant | False | | | The start time.
Slot.status | code | Included only if returned by $find | | | Will be busy-tentative, but only included if returned by $find.

Sample request

```
https://hostname/instance/api/FHIR/STU3/Slot/enA9Zi4LG5esVkLtiqg-
    ZteK8whUj9tp0vDGO9q696Q55fJbgrlQmbWK3sqSzcmSUvlN61UCcZZUARYPUS4lfwg3
```

Sample response

```
<Slot xmlns="http://hl7.org/fhir">
  <id value="enA9Zi4LG5esVkLtiqg-ZteK8whUj9tp0vDGO9q696Q55fJbgrlQmbWK3sqSzcmSUvlN61UCcZZUARYPUS4lfwg3"/>
  <schedule>
    <reference value="https://hostname/instance/api/FHIR/STU3/Schedule/eqhHOcjXjoExdUT.HmVM17Uz5uRvegloDNYibdSkaCu43"/>
  </schedule>
  <start value="2018-07-30T15:30:00Z"/>
  <end value="2018-07-30T16:30:00Z"/>
</Slot>
```

30.3 Patient Workflow Scheduling: FHIR (unversioned)

File: site/apis/30-Patient-Workflow-Scheduling/3-FHIR-unversioned/index.html
* COVID SMART Scheduling Links

30.3.1 COVID SMART Scheduling Links

File: site/apis/30-Patient-Workflow-Scheduling/3-FHIR-unversioned/10061-COVID-SMART-Scheduling-Links.html
Epic API ID | 10061
FHIR version | FHIR (unversioned)
Categories | Patient Workflow Scheduling
API type | Industry-Standard
USCDI | No
Available in inpatient and other contexts | Yes
ONC certification products | EpicCare Ambulatory Base, EpicCare Inpatient Base
ONC certification versions | February 2020
HTTP method | GET
URL template | /api/epic/2021/Scheduling/Utility/CovidVaccineSlotAvailability/{state_code}
Supports OAuth 2.0 | True
Supported user types | Backend Systems and Non-OAuth 2.0, Clinicians or Administrative Users, Patients
Performs scope validation | False
Source | https://fhir.epic.com/Specifications?api=10061

Description

Provides an Epic implementation of the SMART Scheduling Links standard, a lightweight appointment availability publishing API designed to make it easy for organizations to publish COVID vaccine appointment slot availability in cooperation with outside websites, vendors, or government programs.

View the specification here.